Karthik Bhattaram

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Education

University of California, Santa Barbara — Santa Barbara
Master of Science in Computer Science — *March 2026 to March 2027*

- Courses: Advanced Computer Security, Frontier LLM and Agent Capabilities, Graph Neural Networks
- Researching machine learning and cybersecurity under Christopher Kruegel and Giovanni Vigna

University of California, Santa Barbara — Santa Barbara
Bachelor of Science in Computer Science — *September 2022 to March 2026*

- Selected Courses: Data Structures and Algorithms, Operating Systems, Computer Networks
- Honors: Regents Scholar, College of Engineering Honors, Distinction in Major

Experience

Machine Learning Research Assistant — June 2026 – September 2026
Lawrence Berkeley National Lab — *Berkeley, California*

- Developed evaluations to compare network foundation models with traditional deep learning architectures
- Showed that lightweight models could achieve similar performance while using 90% less memory

Security Research Assistant — October 2025 – June 2026
Shellphish — *Santa Barbara, California*

- Built a benchmark of 8 intrusion detection systems against 9,700+ C2 beacon samples spanning 10 frameworks
- Trained a novel detector that achieved best-in-class robustness when samples were adversarially perturbed

Software Engineer Intern — June 2025 – September 2025
Arista Networks — *Santa Clara, California*

- Accelerated memory diagnostics by developing C++ utilities to decode registers in Broadcom's switching ASICs
- Developed an inter-process communication tool for use in OpenAI's datacenters to deliver diagnostic information

Machine Learning Research Intern — September 2024 – June 2025
University of California, Santa Barbara — *Santa Barbara, California*

- Published a paper titled "Leveraging Large Language Models to Contextualize Network Measurements"
- Aggregated FCC data into a SQL database to improve model context using retrieval-augmented generation

Software Engineer Intern — July 2024 – September 2024
Cadence Design Systems — *Santa Fe, New Mexico*

- Wrote AWS-based package to benchmark database compression algorithms that saved terabytes of server space
- Updated CUDA C++ software to incorporate new language features to match competing industry solutions

Hardware Architecture Intern — June 2023 – September 2023
Cadence Design Systems — *San Jose, California*

- Wrote a Python automation tool that reduced memory wrapper development time by over 50%
- Collected foundry provided memories written in System Verilog to be used as templates for this tool

Security Software Intern — June 2021 – August 2022
Axiado Corporation — *San Jose, California*

- Developed a Python package to create network packet flows for use in hardware simulation and emulation
- Created and executed a test plan for a hardware firewall's parser module and discovered critical bugs

Projects

Graph Generation Fine Tuning | *Python, PyTorch, Graph Algorithms* — January 2026 – March 2026

- Adapted Meta's Llama 3.1 model to process graph generation tasks using parameter-efficient fine tuning
- Compared zero-shot, supervised fine tuning, direct preference optimization, and steering performance
- Achieved 85% improvement in graph correctness (based on graph metrics) over the zero shot model

Technical Skills

Certificates: Enterprise Security Professional, Python for Data Science and AI
Languages: Python, C/C++, Java, JavaScript, Shell, Verilog
Developer Tools: Git, Visual Studio Code, Vim, Emacs